

Ian Kaufman

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EDUCATION

Stony Brook University

Expected Graduation: May 2027

Bachelor of Science in Computer Science & Applied Math and Statistics

GPA: 3.64 / 4.00

Relevant Coursework: Software Development, Data Structures, Analysis of Algorithms, Scripting Languages, Object-Oriented Programming, Data Science, Probability & Statistics, Artificial Intelligence, Operating Systems, Computational Theory, Calculus 3

Teaching Assistant: Mentored students in Finite Mathematical Structures and Technology in Media

SKILLS

Languages: Python, C++, SQL, R, JavaScript, HTML/CSS, Linux/UNIX Shell Scripting, Java, C

Frameworks & Libraries: Django, FastAPI, Uvicorn, SQLAlchemy, React, Node.js/Express

Databases & Data Technologies: PostgreSQL, MongoDB, SQLite, BigQuery, Firestore, Supabase

Cloud & Infrastructure: AWS (EC2, S3, Lightsail), Azure App Service, Google Cloud Platform, Docker, GitHub Actions, CI/CD

PROFESSIONAL EXPERIENCE

Amazon Web Services— AWS Training & Certification

Portland, OR

Software Development Engineer Intern

August 2026 - Present

- **Developed an AI-powered Learner Assistant using Amazon Bedrock**, delivering **3 learner-facing capabilities**—Socratic tutoring, personalized post-assessment coaching, and contextual issue reporting—with guardrails preventing users from obtaining assessment answers with localization infrastructure across 3 languages.
- **Redesigned and refactored the AWS Skill Builder Assessment Player**, decomposing thousands of lines of unstructured code into a modular Model-View-Container architecture and rebuilding core assessment workflows with React, TypeScript, and GraphQL to improve maintainability, accessibility, and developer velocity.
- **Contributed to production feature development on 10+ tickets** and bug resolution across AWS Skill Builder, including building environment-aware configuration mappings across **4 deployment stages (development, beta, gamma, production)** to automatically resolve internal identifiers and eliminate manual configuration for future developers.

Ally Financial

Charlotte, NC

Technology Intern—Systems Architecture (Enterprise and Technology)

May 2026 - August 2026

- Designed and developed an end-to-end **Splunk—ServiceNow API service** in an Agile environment that automated firewall request submissions, IP/subnet traffic analysis, flow selection, and access request formatting; **saved 15+ hours weekly**, **reducing manual effort by 30%**, and **accelerating workflows by 25%** according to Platform Automations team estimates.
- Preserved **500 limited Nutanix project categories** and improved metadata visibility by an estimated **60%** by designing a metadata-to-JSON conversion utility that injects structured JSON directly into projects while scaling tagging efficiently.

MyStage Music

Remote

Software Engineering Intern

Jan 2026–May 2026

- Architected and deployed an end-to-end deduplication and automerger pipeline processing **300K+ Firestore records**, using Firebase, Pydantic, BigQuery SQL transformations, and Firestore writeback to surface validated results in the admin UI.
- Containerized the deduplication and automerger pipeline with Docker for **Google Cloud deployment**, enabling weekly runs to refresh validated Firestore records automatically, saving **30+ hours** monthly by replacing manual cleaning with automation.

PROJECTS

Book Recommender | <https://iansbookrecs.com> | Python, FastAPI, JavaScript, scikit-learn, AWS, REST APIs

- Deployed a full-stack book recommendation app on AWS Lightsail using FastAPI, SQLite, Docker, and a JavaScript frontend/Python backend, serving **30+ users** with account-based libraries and personalized recommendations.
- Built **per-user accounts** with **PBKDF2-hashed passwords** and session-cookie authentication, a personal library with user-defined sections and reading statuses driving **scoped recommendations**, and a **thread-safe LRU cache** that skips unchanged recommendation runs received from a **content-based recommendation engine**.

Distributed Task Queue | github.com/Iank314/task-queue | aiohttp • PostgreSQL • Python • Docker Compose • GitHub Actions

- Designed a distributed task queue with an **aiohttp broker**, **PostgreSQL-backed job store**, horizontally scalable worker pool, and row-level locking, preventing duplicate job execution across concurrent workers through atomic job claiming.
- Improved queue reliability and observability across **4+ operational metrics** by implementing priority scheduling, exponential backoff retries, a dead-letter queue, and tracking for queue depth, throughput, job states, and failure rates.

LEADERSHIP

Vice President & Co-Founder

Stony Brook University

Business Analytics Club

Jan 2026 – Present

- **Co-founded and led** a student organization with 30+ members focused on analytics and data-driven decision making.
- Organized industry speaker events with professionals from finance/tech firms such as **Alvarez & Marsal**; 50+ in attendance.